How sharp is our soil-moisture map, really?

A plain-language guide to the effective-resolution test (runbook §70)

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6 October 2026

0. The question in one paragraph

Our model draws a soil-moisture map on a grid of 20 m × 20 m pixels. That does *not* mean the map really knows what happens every 20 m. A photo can have millions of pixels and still be blurry. The **effective resolution** is the size of the smallest detail that is (1) actually present in the map and (2) actually correct. This document explains how we measured both, and what we found.

In one sentence: The map has 20 m pixels, but we want to know the size of the smallest detail that is real.

1 What the model produces

- The world is cut into square **tiles** of 2.24 km × 2.24 km.
- For each tile and each day, the model outputs a map of 112 × 112 pixels (112 × 20 m = 2.24 km). It does this for three depths: 0–10, 10–30 and 30–100 cm.
- Inside the model, each tile is first described by a coarser grid of 14 × 14 **tokens**. Each token covers 160 m × 160 m. These tokens carry most of the satellite information. A separate "fine path" also feeds 20 m satellite pixels into the last stage.
- We test on **TxSON**, a dense sensor network in Texas: 40 soil-moisture stations, many of them only a few hundred metres apart. 30 of the tiles were never used for training ("held-out"). All numbers below come from those 30.

2 Removing the pattern that never changes

Figure 1 shows one real tile. Panel (a) is the *average* map over all days. It has a clear pattern, but earlier work (§56) showed that this pattern is almost the same every day: the model paints it regardless of weather. A fixed pattern tells us nothing about whether the model follows real wetting and drying.

So, for every pixel, we subtract that pixel's own long-term average. What is left is the **anomaly**: "today this spot is wetter (blue) or drier (red) than it usually is" (panel c). *Every measurement in this document is made on anomaly maps.*

In one sentence: We throw away the part of the map that never changes and study only the day-to-day changes.

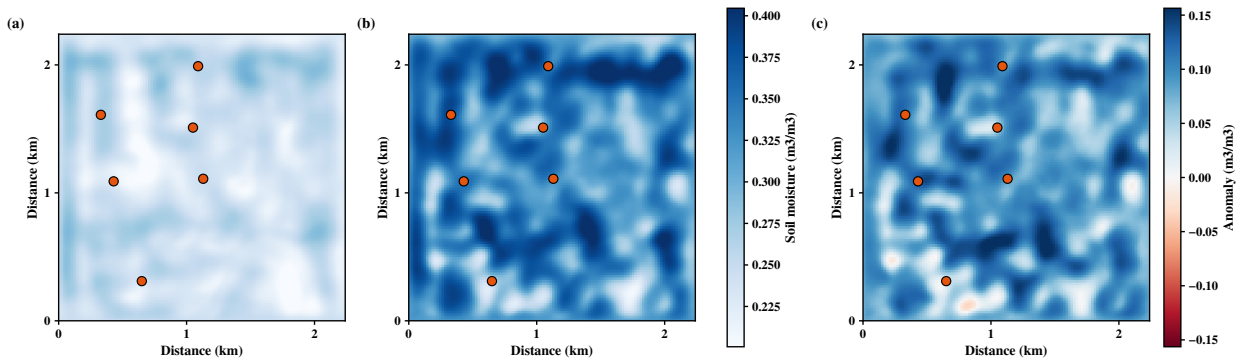


Figure 1: One held-out tile (TxSON CR200-18, 0–10 cm). (a) The average map over all days, which is the fixed pattern. (b) The map on one day (the day with the most spatial contrast). (c) That day minus the average: the anomaly. Orange dots are the real sensor stations inside this tile.

3 Three ways to ask "how fine is the detail?"

All three tools below measure the same property from different angles.

3.1 The variogram: "how different are two points that are h metres apart?"

Take many pairs of points at the same distance h (say 100 m). For each pair, measure how much their values differ, then average. Repeat for h = 20 m, 40 m, …, 1.6 km. The resulting curve is the **variogram**. We divide it by the total variance, so 0 means "identical" and about 1 means "completely unrelated".

Figure 2 shows two made-up maps to build intuition:

- **Smooth field (a):** neighbours look alike. The curve starts at 0 and rises slowly. It reaches its plateau only at several hundred metres. The distance where it flattens is called the **range**: beyond it, points are as different as two random points.
- **Rough field (b):** even neighbouring pixels differ, so the curve *jumps up immediately*. The jump at very short distance is called the **nugget**. It means there is detail finer than we can see, or simply noise.

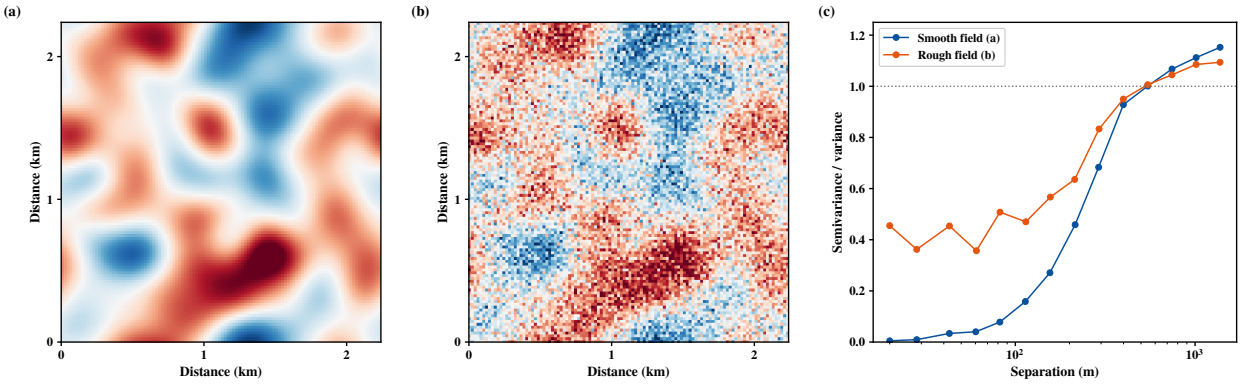


Figure 2: Toy example, not model output. (a) A smooth field with blobs a few hundred metres wide. (b) The same field with 20 m-scale noise added. (c) Their variograms: the smooth field rises gradually and has no nugget; the rough field jumps up at the first lag.

In one sentence: Where the variogram flattens (the range) is the size of the blobs in the map.

3.2 The spectrum: "how much of the map is made of waves of each size?"

Any map can be written as a sum of waves of different lengths: long gentle waves (2 km) plus shorter ripples (40 m). The **spectrum** measures how strong each wave length is. A smooth map is strong in long waves and has almost nothing in short ones. The spectrum and the variogram carry the same information; one speaks in wave lengths, the other in distances.

3.3 The blur test: "what do I lose if I blur the map?"

Average the map over blocks of $b \times b$ metres and measure how much of its variation survives. If blurring at 80 m keeps 100 %, the map had no detail at 80 m. The block size where the variation starts to disappear is where the detail lives.

4 Why 160 m and 320 m are marked on the figure

The model's tokens form a 160 m grid. Figure 3 shows what such a grid can and cannot hold.

- To draw a wave you need at least one wet cell and one dry cell, so at least **two cells per wave**. With 160 m cells, the shortest wave the grid can hold is $2 \times 160 = \mathbf{320\,m}$ (panel a). This limit is called the **Nyquist** limit.
- A wave shorter than that, for example 100 m (panel b), falls *inside* the cells. Each cell only keeps its average, which is close to zero, so the wave is lost.

So if the map's detail comes from the tokens, we *expect* it to stop at around 320 m wave length (160 m in distance). If the 20 m fine path added real detail, we would see power at shorter waves too.

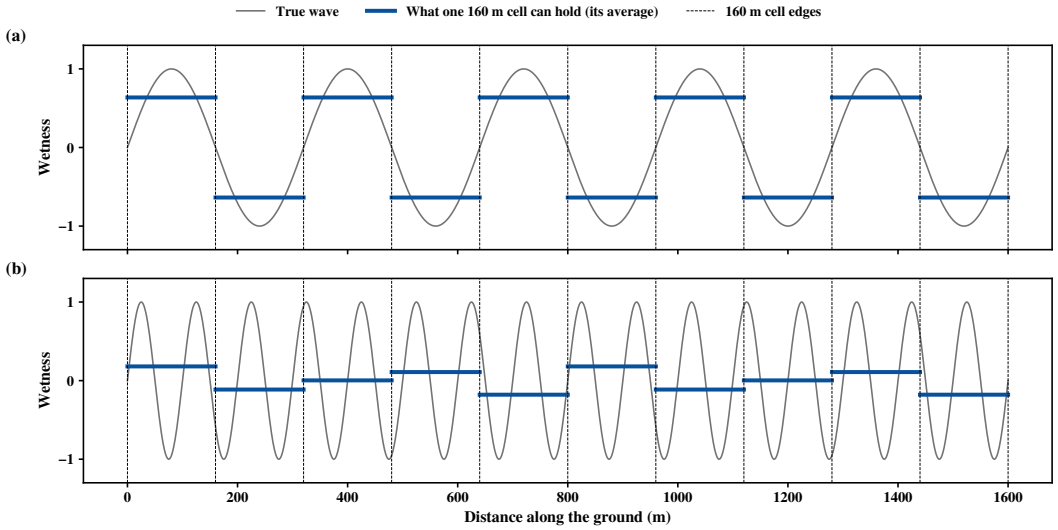


Figure 3: Toy example. (a) A 320 m wave on a 160 m grid: the cell averages (blue) still go up and down, so the grid can carry this wave. (b) A 100 m wave: the cell averages are almost flat, so the grid cannot carry it.

In one sentence: A 160 m grid can only draw waves 320 m long or longer; that is why those two lines are on the plots.

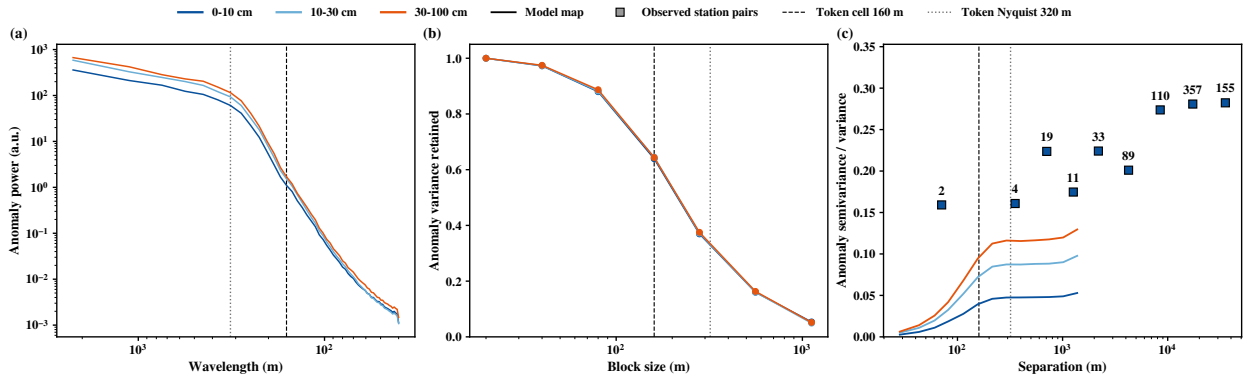


Figure 4: Result over the 30 held-out TxSON tiles. Colours are depths (dark blue 0–10 cm, light blue 10–30 cm, orange 30–100 cm). Dashed line: token cell, 160 m. Dotted line: token Nyquist, 320 m. (a) Spectrum of the anomaly maps. (b) Variation retained after blurring. (c) Variogram: lines are the model map; squares are real station pairs at 0–10 cm, with the number of pairs written above.

5 The result figure, panel by panel

Panel (a), spectrum. Read from left (long waves) to right (short waves). The curve is high and slowly falling down to about 320 m (dotted line), then it **drops about 100 times** by 160 m (dashed line). Below that there is almost nothing. *Meaning:* the map's day-to-day pattern is made of blobs about 300 m and larger, which is what the 160 m token grid can draw. The 20 m fine path adds almost nothing.

Panel (b), blur test. Blurring the map into 40 m blocks keeps 97 % of its variation; 80 m keeps 88 %, 160 m 64 %, 280 m 37 % and 560 m 16 %. *Meaning:* there is almost no detail at 20–80 m; the detail lives between about 160 and 500 m.

Panel (c), variogram.

- **Lines (model):** start near 0 (neighbouring pixels are near-copies), rise until about 250 m, then go flat. So the **range of the map is about 250 m**.
- **Height of the flat part:** 0.05 at 0–10 cm. Within a tile, all points move almost together: only about 5 % of their daily change differs from point to point.
- **Squares (real stations):** 0.16–0.22 at distances below 3 km and about 0.28 at 6–50 km. Real points disagree about **4 times more** than the model's points do. Part of the real difference is sensor noise, so 4× is an upper limit on the gap.

A note on the dotted 320 m line: it is exact only in panel (a), where the axis is a wave length. In panels (b) and (c) the axis is a block size or a distance, so the matching marker there is the dashed 160 m line.

In one sentence: The map contains detail down to about 250 m, and it is much smoother than the real ground.

6 Present is not the same as correct

The figure shows what detail the map *has*. It cannot show whether that detail is *right*. Three extra tests, compared against the real stations, answer that.

In one sentence: Blurring the map to the whole tile does not hurt it, so the detail inside a tile is not skill.

Test	What we did	What happened
C: blur the output	Blur the predicted map around each station, from 20 m up to the whole tile (2.24 km), and recompute the error against the station.	The error **never gets worse**, even with the whole tile blurred to one value (0.0452 $\rightarrow$ 0.0443 m^3/m^3). The fine detail adds no accuracy.
Control	Run the same test on a made-up "perfect" map that truly has 160 m detail.	The test answers 280 m. So the test *can* detect fine resolution when it exists.
B: blur the input	Blur the satellite inputs before they enter the model (40 m to 2.24 km) and predict again.	The map **changes a lot** (30 % of its spatial variation already at 40 m, 58 % at 160 m), but the station error **does not change**. The fine inputs reshape the picture without making it more accurate.
A: station pairs	For pairs of stations, compare the predicted difference with the observed difference.	Long-term differences: correlation about 0. Day-to-day differences: a weak but real signal at 500–1000 m (correlation 0.08, 95 % range 0.02–0.11).

Table 1: The correctness tests (0–10 cm only; deeper layers have too few station pairs).

7 Conclusion

1. The map **contains** detail down to about **250 m**. That is set by the model's 160 m token grid, not by the 20 m output pixels.
2. The detail is **not correct**. Removing it entirely does not make the predictions worse. The **effective resolution is about one tile, 2 km**, at 0–10 cm.
3. Within a tile the model draws **texture**: the inputs move it around, but it does not follow the real differences between nearby points. It is also about 4 times smoother than the real ground.

8 How every number is computed

This section lists the exact recipe behind each curve and test. Code: `effres_maps.py` (GPU, makes the maps), `effres_analysis.py` (Tests 0, A, B, C, control, spectrum, blur test) and `plot_effres.py` (model variogram, figure). "Held-out" always means the 30 TxSON tiles and stations the model never trained on.

8.1 Common first step: the anomaly maps

1. Run the trained model once on every day of every tile: about 2,500 days $\times$ 3 depths $\times$ 112×112 pixels per tile.
2. For each pixel, take its average over all days (the fixed pattern, Figure 1a).
3. Anomaly = map of the day $-$ that average, pixel by pixel (Figure 1c).

8.2 Anomaly power (Figure 4a, the spectrum)

For each tile and depth:

1. **Remove the tile-wide shift of each day.** Subtract each day's spatial mean from its anomaly map. On a wet day the whole tile is, say, $+0.1$; that uniform part is dropped, so only the differences *within* the tile remain.
2. **Split the map into waves.** A 2-D Fourier transform (FFT) rewrites the 112×112 map as a sum of 2-D sine waves. Each wave has a direction, a strength and a frequency k: the number of full waves that fit across the tile ($k = 1 \ldots 56$).
3. **Power = strength squared**, for every wave. A large value means the map contains a lot of that wave.
4. **Average over days.**
5. **Pool all directions.** A 300 m blob is the same size whether it runs north–south or east–west, so all waves with the same k are averaged together whatever their direction (they lie on a ring of radius k in the Fourier plane). This leaves one number per k; hence the name *radially averaged* spectrum.
6. **Convert k to a wave length:** wave length $= 2240\,\mathrm{m}/k$. So $k = 1$ is 2240 m (one wave across the tile), $k = 7$ is 320 m (the shortest wave a 160 m token grid can carry), $k = 14$ is 160 m, and $k = 56$ is 40 m (the shortest wave a 20 m pixel grid can carry).
7. **Combine tiles:** the median over the 30 held-out tiles at each k. This is the plotted curve.

Why "a.u." (arbitrary units)? The FFT is not normalised, so the absolute numbers depend on the map size. Only the *shape* and the *ratios* matter. Summed over all k, the power is proportional to the spatial variance of the anomaly map, so the curve says how that variance is shared out among wave sizes.

8.3 Variance retained (Figure 4b, the blur test)

For each tile, depth and block size b (40, 80, 160, 280, 560, 1120 m):

1. Cut each day's anomaly map into $b \times b$ blocks (for 160 m: 14×14 blocks of 8×8 pixels) and replace each block by its mean.
2. Measure the spatial variance of the blurred map (variance across blocks) and of the original map (variance across pixels). A variance across space ignores the tile-wide shift automatically.
3. Average each variance over all days, then take the ratio: blurred $\div$ original.

4. Median over the 30 tiles. 1.0 means blurring lost nothing; 0.0 means everything was smaller than the block.

8.4 Model variogram (Figure 4c, lines)

For each tile and depth:

1. Pick 40,000 random pairs of pixels in the tile. For each pair, note its distance h (in metres).
2. For that pair, take the two anomaly time series a_t and b_t and compute
   - the **semivariance** $g = \frac{1}{2} \cdot \text{mean}_t\,(a_t - b_t)^2$, which measures how differently the two points behave over time;
   - the **variance** $v = \frac{1}{2}\,(\text{var}_t\, a + \text{var}_t\, b)$, which measures how much each point changes over time on its own.

   A tile-wide shift on one day moves a_t and b_t equally, so it cancels in $a_t - b_t$.
3. Group the pairs into 14 distance bins (logarithmic, 20 m to 1.6 km). Average g and v in each bin.
4. Average g and v over the 30 tiles, then plot $g \div v$ against distance. 0 means the two points move identically; 1 means as different as two unrelated points.

8.5 Observed variogram (Figure 4c, squares; "Test 0")

1. Take every pair of the 40 TxSON stations, using the days on which both measured (at least 60 days in common).
2. Anomaly of each station = its value − its own mean over those common days. This matches the model's per-pixel average in 8.1.
3. Compute g and v exactly as in 8.4. The distance comes from the stations' latitude and longitude.
4. Group the pairs into distance bins (0–250 m, 250–500 m, 0.5–1 km, 1–1.6 km, 1.6–3 km, 3–6 km, 6–12 km, 12–25 km, 25–50 km). Plot (mean g) ÷ (mean v). The number above each square is the number of station pairs.

Same formula as the model, so the two can be compared directly. Two cautions: a sensor measures a point, not a 20 m pixel, and sensor noise adds to g. So the observed values are an upper limit for the real spatial variation.

8.6 Test C: does blurring the output hurt accuracy?

1. Use each held-out tile whose own station sits at the centre pixel. That gives 30 tiles.
2. For each block size b (20 m … 2.24 km), average the predicted map over a $b \times b$ window around the station on every day. This gives one predicted time series per b.
3. Error = **ubRMSE** against the station: the root-mean-square difference after removing each series' own mean.
4. Δ = ubRMSE at b − ubRMSE at 20 m, per tile, then averaged over tiles.
5. 95 % interval: resample the 30 tiles with replacement 1,000 times (a bootstrap). A block is "significantly worse" if the whole interval is above 0.
6. L_C = the largest block that is not significantly worse. Result: 2240 m, because no block was worse.

8.7 Control: can Test C detect real fine detail?

1. Make 30 synthetic tiles of 300 days each. The "true" field is random noise smoothed to blobs of about 160 m (Gaussian, $\sigma = 160$ m).
2. "Perfect model" = the true field plus a little 20 m noise. "Station" = the true field at the centre pixel plus a little sensor noise.
3. Run exactly the Test C recipe. Result: blocks up to 280 m are fine, 560 m and larger are significantly worse, so $L_C = 280$ m. The test works; the real model simply has no fine skill to find.

8.8 Test B: does the model use fine inputs?

1. For each block size b, blur the tile's *inputs* before they enter the model: the 20 m satellite channels (S2, S1, DEM) are block-averaged using valid pixels only; land cover takes the most common class in each block; the soil map is block-averaged; and the 14×14 tokens are block-averaged once b exceeds 160 m.
2. Run the model again on every day (one full GPU pass per b).
3. Map change S = RMS(new map $-$ original map) $\div$ spatial SD of the original map, per day, then the median. $S = 0$: blurring the inputs changed nothing. $S \approx 1$: the change is as large as the map's whole spatial pattern.
4. Also recompute the station ubRMSE at the centre pixel with the blurred inputs.
5. L_B = the largest b with $S < 0.1$. There is none: S is already 0.29 at 40 m. Yet the ubRMSE stays at 0.044–0.045 for every b.

8.9 Test A: are predicted differences between stations right?

1. For a pair of held-out stations A and B: predicted difference $d_t^{\text{pred}} = \text{pred}_A - \text{pred}_B$, and observed difference $d_t^{\text{obs}} = \text{obs}_A - \text{obs}_B$, over their common days. *Within-tile pairs* use the station pixels of one map (same model run). *Cross-tile pairs* use the centre pixel of each station's own tile.
2. **Level skill:** the long-term mean of d^{pred} versus that of d^{obs}, correlated across all pairs in a distance bin. This asks: is the wetter station predicted wetter?
3. **Dynamic skill:** the correlation over time between d_t^{pred} and d_t^{obs} for each pair, then the median over pairs in the bin. This asks: when A gets wetter than B, does the model see it?
4. 95 % intervals: a bootstrap that resamples whole tiles (or stations, for cross-tile pairs), so that pairs sharing a tile are not counted as independent.
5. L_A = the shortest distance bin where either interval is entirely above 0. Result: 765 m (the 0.5–1 km bin, dynamic $r = 0.08$).

8.10 The verdict

Effective resolution L_{eff} = the largest of L_A, L_B and L_C, which is the most demanding answer. At 0–10 cm: $\max(765, \text{none}, 2240) = \mathbf{2240\ m}$.

Glossary

Anomaly A pixel's value today minus its own long-term average.

Effective resolution
: The size of the smallest detail that is both present and correct.

Nugget
: The jump of a variogram at very short distance: detail finer than the grid, or noise.

Nyquist limit
: The shortest wave a grid can carry: two grid cells.

Range
: The distance where a variogram flattens; roughly the size of the blobs in the map.

Spectrum
: How strong each wave length is in a map.

Token
: One 160 m $\times$ 160 m cell of the model's internal 14×14 description of a tile.

ubRMSE
: The typical prediction error after removing the constant offset between model and station.

Variogram
: Half the average squared difference between two points, as a function of their distance.